

- There are 5 Questions for 25 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof).
- You do not need any calculators.
- Good luck!

- [5] Let $f : A \rightarrow B$. For $X \subseteq A$ and $Y \subseteq B$, let $\hat{f}(X) = \{f(x) \mid x \in X\}$. Prove that for all $X_1, X_2 \subseteq A$:
 - $\hat{f}(X_1 \cup X_2) = \hat{f}(X_1) \cup \hat{f}(X_2)$
 - $\hat{f}(X_1 \cap X_2) \subseteq \hat{f}(X_1) \cap \hat{f}(X_2)$, but equality need not hold.
- [5] In each of the following statements, if we replace S by one of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, then we get a true statement. Find the appropriate set in each case:
 - $\{x \in S \mid 1 < x \leq 4\} = \{x \in S \mid x^2 = 4\} \cup \{3, 4\}$
 - $\{x \in S \mid 4x^2 = 1\} \setminus \{x \in S \mid x < 0\} = \{x \in S \mid 5x^2 = 3\} \cup \{x \in S \mid 2x = 1\} \neq \emptyset$.
- [5] Given a vector $\underline{x}$, we define the difference operator ∇ as

$$\nabla \underline{x} = \begin{pmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_4 - x_3 \\ \vdots \\ x_n - x_{n-1} \end{pmatrix} \text{ where } \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Is ∇ linear? If yes, what is the matrix representation of ∇ ? Is ∇ one-one?

- [5] Suppose $f : A \mapsto B$ is right invertible. Show that f is onto.
- [5] Suppose f is a degree 2 polynomial in a single variable,

$$f(x) = a_0 + a_1x + a_2x^2$$

which satisfies the following

$$f(x_1) = r_1, \quad f(x_2) = r_2, \quad f(x_3) = r_3.$$

We are given x_1, x_2, x_3 and r_1, r_2, r_3 , and we want to find f , i.e. a_1, a_2, a_3 . Suppose x_1, x_2, x_3 are fixed; and we have

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = g \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix},$$

i.e. g is the map from the values r_1, r_2, r_3 to the polynomial coefficients a_1, a_2, a_3 . Is g linear? If yes, What is the matrix representation of g ?